Predictive Movement Modeling Using HMMs and DBNs

Enhancing Automation Through Context-Aware Prediction

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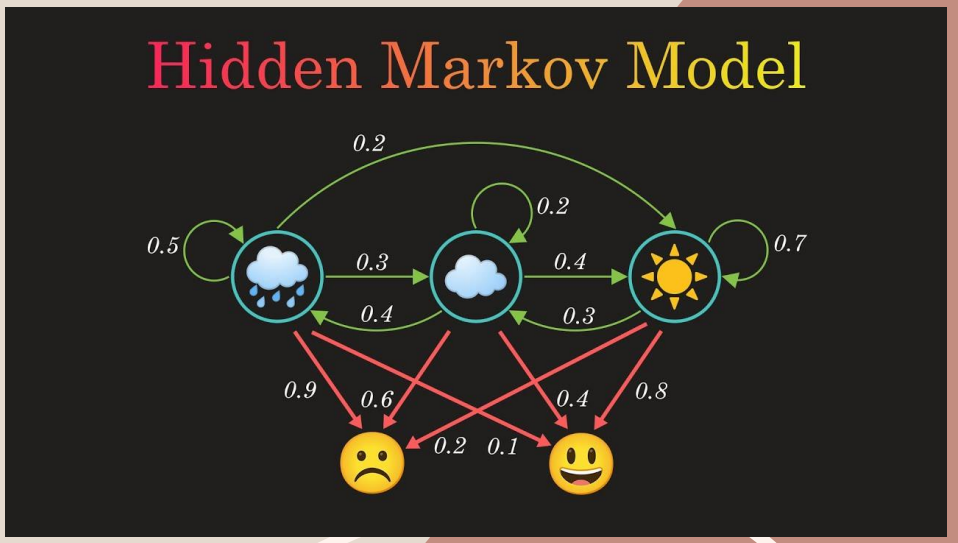
Introduction & Motivation

- Problem:
  Smart homes require proactive automation (lighting, security, energy efficiency).
- Challenges:
  Complex user behavior, irregular routines, and periodic patterns.
- Solution:
  Predict user movements to enable context-aware automation.
- Impact:
  Energy savings, enhanced security, elderly care support.

Objective

Goal:
Predict a user's next location in a smart home using historical sensor data and contextual features (e.g., time of day).

Key Focus:
Hybrid model combining Hidden Markov Models (temporal patterns) and Dynamic Bayesian Networks (contextual dependencies).



Why HMMs and DBNs?

HMMs:

- Ideal for temporal sequences (e.g., room transitions).
- Computationally efficient, interpretable.

DBNs:

- Extend HMMs by integrating context (time of day, habits).
- Better accuracy for periodic routines.

Comparison with Alternatives:

- LSTMs/GRUs require more data; HMMs/DBNs balance efficiency and accuracy.

Dataset & Preprocessing

Dataset: CASAS Aruba

- Motion sensor activations (e.g., "Bedroom," "Kitchen").
- Temporal sequences with ground-truth labels.

Preprocessing Steps:

- Split data into training (70%) and testing (30%).
- Map sensor IDs to room labels.
- Add time-of-day context (morning/afternoon/night).

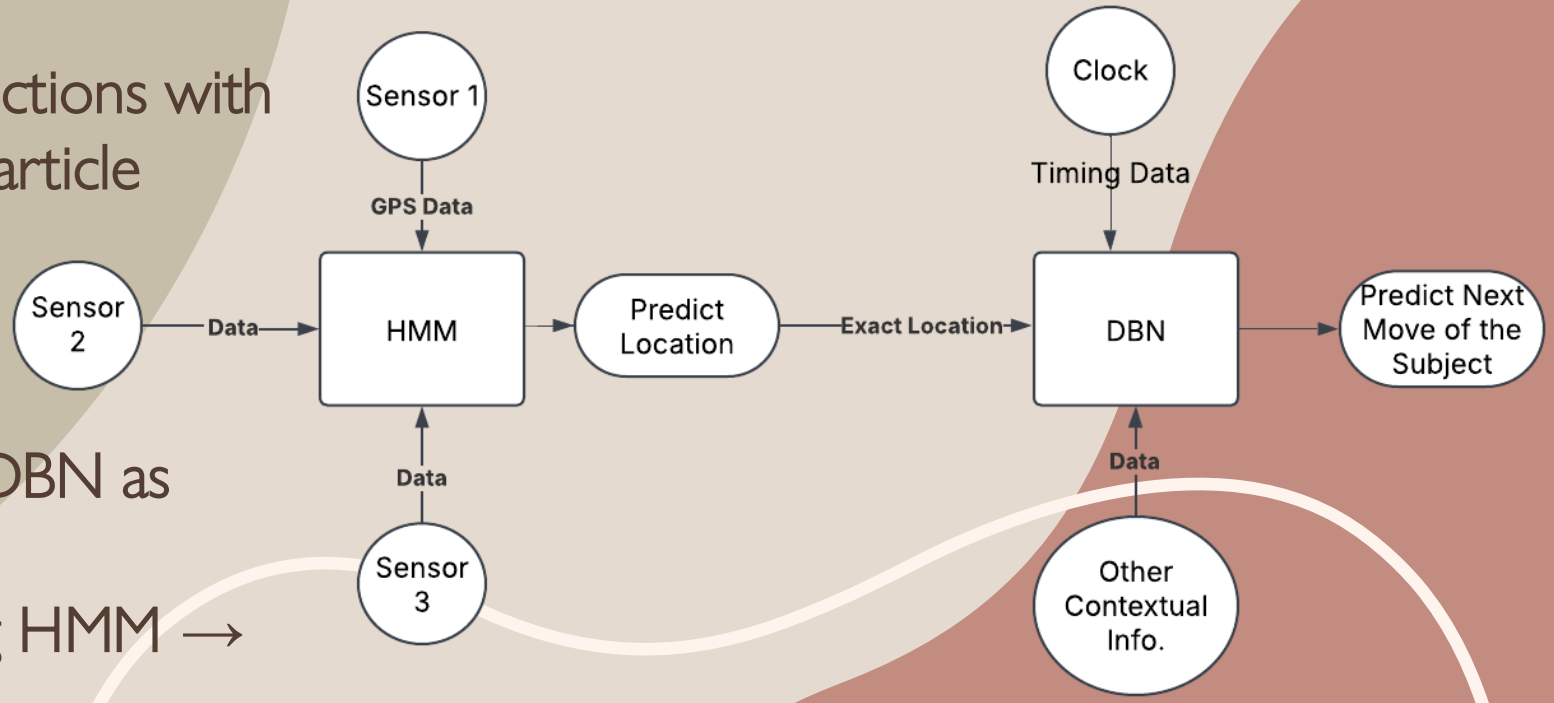
Model Design

Hybrid Architecture:

- HMM Layer: Predicts room transitions using Viterbi algorithm.
- DBN Layer: Refines predictions with time-of-day context via particle filtering.

Integration:

- HMM output feeds into DBN as contextual input.
- Visual: Flowchart showing HMM → DBN interaction.



Workflow Overview

1. Data Collection: CASAS Aruba dataset.
2. Feature Engineering: Room labels + time bins.
3. Model Training:
    - HMM: Baum-Welch algorithm.
    - DBN: Particle filtering with HMM context.
1. Evaluation: Accuracy, F1-score, confusion matrices.
2. Deployment: Real-time integration with smart home systems.

Evaluation Metrics

1. Accuracy: % of correct room predictions.
2. Precision/Recall: Per-room performance.
3. Transition Confusion Matrix: Common errors (e.g., Living Room vs. Dining Room).
4. Prediction Lead Time: Steps ahead forecasted accurately.
5. Visual: Comparison table (HMM vs. DBN expected accuracy).

Expected Results

HMM Baseline: ~80–85% accuracy.
DBN Improvement: ~85–90% accuracy (better time-context handling).
Visual: Bar chart showing accuracy gains.

Challenges & Solutions

Data Sparsity:

Solution: Laplace smoothing, synthetic sequences.

Overfitting in HMMs:

Solution: Regularization (AIC/BIC).

Temporal Alignment:

Solution: Time discretization (e.g., 6 AM–9 AM = "morning").

Conclusion

Outcome:

- Robust model for room transition prediction.
- Foundation for real-time smart home automation.

Future Work:

- Actuator integration (lights, HVAC).
- Expand to multi-user environments.

Thank You